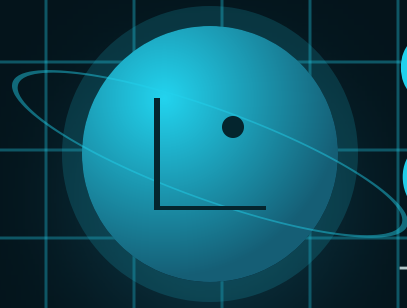




Coordinate Geometry

Two numbers, one place — and a graph that tells the truth.

THE BIG QUESTION

Why does $(3, 5)$ land somewhere completely different from $(5, 3)$?

HANDS-ON PROJECT

Chart the Year

Plot your own scores across all eight missions. One graph, honest axes, and a paragraph on what it shows.

FLUENCY GAME

Treasure Grid

Hide a point, call coordinates, get told hotter or colder. Fewest guesses wins.



AXIOM • MISSION COMMANDER, FINAL TOUR

“Across the hall, then up the stairs. Always in that order — and if you swap them, you end up in somebody else's room.”

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

This is not a test. Nothing here counts for a grade. Write coordinates in the form 3,5 — the across number first.

STRAND A READING A POINT

A1 In (3, 5), which number is across?

.....

A2 Write the origin as a,b

.....

STRAND B DISTANCE

B1 Distance from (2,3) to (7,3)

.....

B2 Distance from (4,1) to (4,8)

.....

STRAND C SHAPES ON A GRID

C1 Area of the rectangle (1,1),(1,5),(6,5),
(6,1)

.....

C2 The missing corner of a rectangle at
(1,1),(1,4),(6,4)

.....

STRAND D READING A GRAPH

D1 An axis marked every 4 — the value 3
marks up

.....

D2 An axis marked every 25 — the value
3 marks up

.....

STRAND E PATTERNS AS POINTS

E1 A rule of $\times 2$ — what does 5 become?

.....

E2 A rule of $\times 2 + 1$ — what does 5
become?

.....

STRAND F ORDER MATTERS

F1 Are (3,5) and (5,3) the same place?
Yes or no

.....

F2 Say why, in one sentence

.....

Skip Chart

Score the pre-assessment, then cross days off the four-week calendar.

2 / 2 → SKIP
1 / 2 → COMPRESS
0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A • Reading a point	A1 A2	Week 1 • lesson 1.1	___ / 2	Day 1.1. This is Year One Mission 06 revisited.
B • Distance	B1 B2	Week 1 • lesson 1.2	___ / 2	Day 1.2, but keep the Challenge tier — diagonal routes are new this year.
C • Shapes on a grid	C1 C2	Week 1 • lessons 1.3 and 1.4	___ / 2	Compress Week 1 to three days.
D • Reading a graph	D1 D2	Week 2 • lessons 2.1 and 2.2	___ / 2	Day 2.1 only. Keep 2.2 — choosing a scale is harder than reading one.
E • Patterns as points	E1 E2	Week 3 • lessons 3.1 and 3.2	___ / 2	Nothing. Computing a rule and seeing it as a line are different skills.
F • Order matters	F1 F2	Week 1 • lesson 1.1 and Week 4 • lesson 4.2	___ / 2	Nothing. This is the Big Question — teach it even at 2/2.

ONE NOTE BEFORE YOU START

This is the last mission of the year, and the project asks him to plot his own scores across all eight. Expect that to be the most useful hour of the mission — a student who can see his own improvement as a line has learnt something about graphs that no worksheet delivers. Have the eight test scores to hand before Week 4 starts.

Across, Then Up

Two numbers name one place. The order is not negotiable.

Warm-Up NAME THE ACROSS NUMBER

(3, 5)

(7, 1)

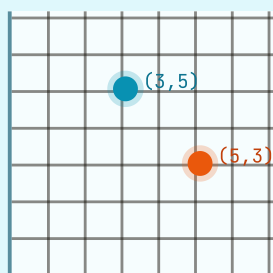
(0, 6)

(4, 0)

(2, 9)

(6, 6)

(3, 5) AND (5, 3) ARE TWO DIFFERENT ROOMS



Both points use a 3 and a 5, and they are nowhere near each other.

The first number is always how far **across**. The second is always how far **up**. Say it out loud as you plot — “three across, five up” — and the mistake becomes impossible to make.

The origin, where the two lines meet, is (0, 0): no distance across, no distance up.

Core WRITE ANSWERS AS A, B

1. 4 across, 2 up

.....

2. 3 right of (2,5)

.....

3. 2 up from (4,1)

.....

4. Swap the two numbers in (7,2)

.....

5. A point sits on the bottom line of the grid. Which of its two numbers must be 0? Say why.

.....

Challenge NOT GRADED

C1. How many points from (0,0) to (10,10) have their two numbers equal? List a few and say what shape they make.

.....

C2. Which points stay in the same place when you swap their two numbers? Describe them all.

.....



AXIOM: Across the hall, then up the stairs. Say it every time you plot a point, and you will never swap them.

How Far Apart

Same row or same column? Subtract. Otherwise, go across then up.

Warm-Up DISTANCES ALONG A LINE

(2,3)→(7,3)

(4,1)→(4,8)

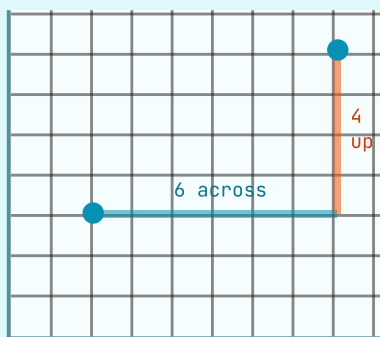
(0,0)→(6,0)

(1,2)→(1,9)

(3,5)→(9,5)

(5,2)→(5,2)

(2,3) TO (8,7) · THE ROUTE IS 10, THE GAP IS NOT



Going across and up, the journey is $6 + 4 = 10$ units. That is the distance if you have to follow streets.

The straight-line gap is shorter than 10 — you can see that on the grid. Finding it exactly needs maths you will meet in a couple of years, so this year we always say which kind of distance we mean.

When both points share a row or a column, there is no difference: just subtract.

Core ACROSS-AND-UP DISTANCE

1. (2,3) to (7,3)

.....

2. (4,1) to (4,8)

.....

3. (2,3) to (8,7)

.....

4. (0,0) to (6,8)

.....

5. Halfway between (2,2) and (8,6) — write the point. How did you find each number?

.....

★ Challenge

C1. Is the across-and-up distance from (2,3) to (8,7) the same whichever order you travel? Try both and say why.

.....

C2. Name two points exactly 5 units apart that share neither a row nor a column, measuring across-and-up.

.....

Shapes From Points

Plot the corners, then read the perimeter and area straight off the grid.

✦ Warm-Up SIDE LENGTHS

(1,1)→(5,1)

(1,1)→(1,4)

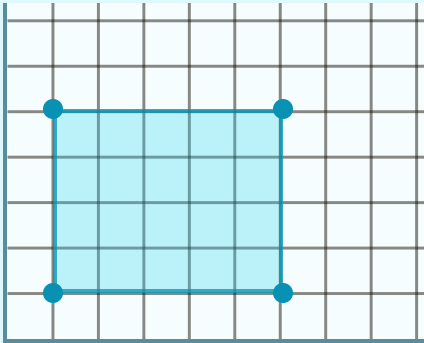
corners on a rectangle

sides on a square

area 5×4

perimeter 5 by 4

FOUR CORNERS, AND EVERYTHING FOLLOWS



Corners at (1,1), (6,1), (6,5) and (1,5).

Width: $6 - 1 = 5$.

Height: $5 - 1 = 4$.

Area: $5 \times 4 = 20$.

Perimeter: $5 + 4 + 5 + 4 = 18$.

No measuring, no ruler — just subtraction on the coordinates.

◆ Core

1. Area of (1,1),(1,5),(6,5),(6,1)

.....

2. Its perimeter

.....

3. Area of (2,2),(2,7),(5,7),(5,2)

.....

4. Area of (0,0),(8,0),(8,5),(0,5)

.....

5. Three corners of a rectangle are (1,1), (1,4) and (6,4). Where is the fourth?

.....

★ Challenge

C1. A square has opposite corners at (2,2) and (7,7). What is its area?

.....

C2. A triangle has corners (1,4), (4,1) and (1,1). Find its area, and say how you handled the sloped side.

.....

Directions Somebody Can Follow

Write a route. Hand it over. Ambiguity is the enemy.

♦ Warm-Up ONE MOVE AT A TIME

(0,0) move 3
across

(3,0) move 2 up

(5,5) move 1
across

(2,2) move 2 up

blocks
(0,0)→(4,0)

blocks
(0,0)→(0,7)

TWO SETS OF DIRECTIONS TO THE SAME PLACE

USELESS

"Go along a bit, then head up past the park until you get near the school."

USABLE

"Start at (1,1). Go 6 across to (7,1). Go 5 up to (7,6). Total: 11 blocks."

Coordinates exist because words like "a bit" and "near" cannot be followed by somebody who is not already there. Today's test is whether your directions work when somebody else reads them.

♦ Core

1. (1,2) to (6,2) then 3 up — the end point

.....

2. Total blocks on that route

.....

3. (1,1) to (7,1) to (7,6) — total blocks

.....

4. Shortest across-and-up route (2,3) to (9,8)

.....

5. Write directions from (0,0) to (5,4) using only coordinates and numbers of blocks. Give them to somebody and see if they land in the right place.

.....

★ Challenge

C1. A patrol goes (1,1) → (7,1) → (7,5) → (1,5) → back to start. Total distance, and the area enclosed?

.....

C2. How many different shortest across-and-up routes go from (0,0) to (2,2)? Draw them all.

.....

Treasure Grid

Hide a point. Call coordinates. Hotter or colder. Fewest guesses wins.

How to play: each player secretly writes a point on a 10 by 10 grid. Take turns calling coordinates. The hider says “hotter” or “colder” compared with the last guess — nothing else. Fewest guesses to find the treasure wins. Guesses must be stated as proper coordinates; “about there” loses the turn.

◆ Your grid

#	GUESS	HOTTER / COLDER
1		
2		
3		
4		
5		
6		
7		
8		

★ Challenge

C1. What is the best opening guess on a 10 by 10 grid, and why? Think about what “colder” rules out.

.....

C2. After three guesses you know the treasure is hotter near (7,7) than (3,3). Which half of the grid can you stop searching?

.....

Weeks 2–4

Every day below has five graded sets waiting in Practice Bay.

QUIZ • end of week 2

TEST • end of week 4

Week 2 • Graphs That Tell the Truth QUIZ FRIDAY

Choosing a scale, and seeing how a dishonest axis makes a tiny gap look enormous. This is Year One's Data Detective, sharpened.

MON 2.1

Read the scale

Marked every 4, every 25 — count before you read.

TUE 2.2

Choose a scale

Data from 0 to 400 on twenty squares — what fits?

WED 2.3

Start the axis at zero

Why starting at 90 is a lie told with true numbers.

THU 2.4

Line graphs over time

When joining the dots is fair, and when it is not.

FRI

Mid-unit quiz

Page 10. Eight items, 85% to keep flying.

Week 3 • Patterns Become Lines

A rule makes a table, a table makes points, and the points fall in a straight line. This is the bridge to algebra.

MON 3.1

Rule to table

A rule of $\times 2 + 1$, worked for 1 to 6.

TUE 3.2

Table to points

Each row becomes a coordinate pair.

WED 3.3

The points make a line

Straight, always — and steeper when the rule multiplies by more.

THU 3.4

Two rules, two lines

Where they cross, and what that point means.

FRI

Chart the Year begins

Collect all eight mission scores.

Week 4 • Proof, and the Year Closes FINAL TEST

The year's own data gets plotted. The final trophy band is awarded at the end of the week.

MON 4.1

Plot eight points

Mission number across, score up.

TUE 4.2

Honest axes

Draw it twice — from 0 and from 60 — and compare.

WED 4.3

Write the paragraph

What the graph shows, and one thing it cannot.

THU

Year-end sweep

All eight error journals. Name the habit that fixed itself.

FRI

Mission 08 test

Pages 11–12. Final trophy band awarded.

Mid-Unit Quiz

Eight items from Weeks 1 and 2. 85% to keep flying — that's 7 out of 8.

1. Write "4 across, 2 up" as coordinates

.....

2. Distance from (2,3) to (7,3)

.....

3. Across-and-up distance from (2,3) to (8,7)

.....

4. Area of (1,1),(1,5),(6,5),(6,1)

.....

5. Halfway between (2,2) and (8,6)

.....

6. An axis marked every 25 — the value 3 marks up

.....

7. Bars at 95 and 98 drawn from an axis starting at 90 — how many times taller does 98 look?

.....

8. A graph starts its axis at 90 instead of 0. Every number on it is true. Explain how it still misleads.

.....

.....

Marking note: item 8 wants the distinction between accurate numbers and a misleading picture. Full marks for any answer saying the bar heights no longer stand in proportion to the values.

Unit Test • Part 1

Items 1–8. No time limit. Write coordinates as a,b.

1. 3 right of (2,5)

.....

2. Distance from (4,1) to (4,8)

.....

3. Across-and-up distance from (0,0) to (6,8)

.....

4. Perimeter of (1,1),(1,5),(6,5),(6,1)

.....

5. The fourth corner of a rectangle at (1,1),(1,4),(6,4)

.....

6. A rule of $\times 2 + 1$ — what does 6 become?

.....

7. That rule turns 6 into 13. Write it as a coordinate pair.

.....

8. A square with opposite corners (2,2) and (7,7) — its area

.....

Unit Test • Part 2

Items 9–12 and the Big Question.

9. A patrol (1,1),(7,1),(7,5),(1,5) and back — the area enclosed

10. An axis marked every 4 — the value 3 marks up

11. A rule of $\times 3$ gives the points (1,3), (2,6), (3,9). Plot them and say what shape they make. What would (5, ?) be?

12. Write directions from (1,2) to (8,6) that somebody else could follow exactly. State the total number of blocks.

EXPLAIN YOUR THINKING • THE BIG QUESTION

Why does (3, 5) land somewhere completely different from (5, 3)?

Sketch both points and say what each number does. Scored on reasoning, not neatness.

11–12

Gold band. The Navigator Badge, and the year's final trophy.

9–10

Silver band. One reteach day, then the badge.

≤ 8

Bronze. Reteach the weak strand and retest that part only.

Key A

Pre-assessment, Lesson 1.1 and Lesson 1.2.

Mark the reasoning, not the handwriting. Any correct method counts.

Pre-assessment

A1 • 3 | A2 • 0,0 | B1 • 5 | B2 • 7
 C1 • 20 | C2 • 6,1 | D1 • 12 | D2 • 75
 E1 • 10 | E2 • 11 | F1 • no | F2 • see below

F2 • because the first number is the distance across and the second is the distance up, so swapping them names a different place. Accept any wording carrying that idea.

Lesson 1.1 • Across, Then Up

Warm-Up • 3 | 7 | 0 | 4 | 2 | 6
 Core • 1 • 4,2 | 2 • 5,5 | 3 • 4,3 | 4 • 2,7

Item 5 • the up number, because sitting on the bottom line means no height at all. C1 • eleven points, from (0,0) to (10,10), and they form a straight diagonal line. C2 • only the points where both numbers are equal.

Lesson 1.2 • How Far Apart

Warm-Up • 5 | 7 | 6 | 7 | 6 | 0
 Core • 1 • 5 | 2 • 7 | 3 • 10 | 4 • 14 | 5 • 5,4

Item 5 • halfway between 2 and 8 is 5, halfway between 2 and 6 is 4. C1 • yes — $6 + 4$ and $4 + 6$ both give 10, since addition does not care about order. C2 • e.g. (1,1) and (3,4), or (2,2) and (5,4): any pair whose across gap plus up gap totals 5.

Key B

Lesson 1.3, Lesson 1.4 and Friday's game.

Where a question asks for a sentence, no sentence means no mark — even with the number right.

Lesson 1.3 • Shapes From Points

Warm-Up • 4 | 3 | 4 | 4 | 20 | 18

Core • 1 • 20 | 2 • 18 | 3 • 15 | 4 • 40 | 5 • 6,1

C1 • 25, since the square has side 5. C2 • 4.5 — half of the 3 by 3 square, and worth discussing that a sloped side does not stop you finding the area.

Lesson 1.4 • Directions Somebody Can Follow

Warm-Up • 3,0 | 3,2 | 6,5 | 2,4 | 4 | 7

Core • 1 • 6,5 | 2 • 8 | 3 • 11 | 4 • 12

Item 5 • the real test is whether somebody else lands on (5,4); a route that needs explaining does not count. C1 • 20 units travelled, enclosing an area of 24. C2 • six routes, and drawing all six is the whole exercise.

Friday • Treasure Grid

C1 • near the middle, around (5,5), because the first “colder” then rules out roughly half the grid whichever way it falls. C2 • the half nearer (3,3) — the lower-left region can be abandoned. Both questions are really about halving the search space, which is a genuinely useful idea and worth naming out loud.

Key C

Mid-unit quiz, final test, and the error journal sheet.

Quiz is out of 8, test out of 12. 85% is 7/8 and 11/12.

Mid-unit quiz

- 1 • 4, 2
- 2 • 5
- 3 • 10
- 4 • 20
- 5 • 5, 4
- 6 • 75
- 7 • 1.6
- 8 • see below

Final test

- 1 • 5, 5
- 2 • 7
- 3 • 14
- 4 • 18
- 5 • 6, 1
- 6 • 13
- 7 • 6, 13
- 8 • 25
- 9 • 24
- 10 • 12

Quiz 8, items 11, 12 and the Big Question

Quiz 8 • the numbers are accurate but the bar heights no longer stand in proportion — a 3% difference is drawn as though it were 60%. **11** • the points form a straight line, and (5,15) continues it. **12** • e.g. "start at (1,2), go 7 across to (8,2), go 4 up to (8,6)" — 11 blocks; accept the up-then-across version. **Big Question** • full marks for a sketch or explanation identifying the first number as distance across and the second as distance up, so that swapping them describes a different location. Credit noticing that the two points are only the same when both numbers are equal.

Error journal • Mission 08 and the year

ONE LINE PER MISTAKE • STUDENT PAGE

Write what went wrong, not what the right answer was. On Thursday of Week 4, read all eight missions' journals and name the one habit that fixed itself over the year.

DAY WHAT I GOT WRONG, AND WHY
